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A Reservoir-Wellbore-Pipeline Integration Method for Gas Lift Performance Optimization Based on Proxy Model and NSGA-II Algorithm of Well Group

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Abstract

This study addresses the complex optimization of gas lift well groups supplied by a single compressor unit under fixed total gas volume constraints. The interdependence of well performance and reservoir management requirements-including reservoir pressure limits, individual production quotas, surface facility constraints, and completion conditions-necessitates a fast-response multi-objective optimization framework. To overcome computational bottlenecks of traditional reservoir modeling, a data-driven proxy model accelerates production prediction by learning from full-field reservoir simulations segmented into well-group performance datasets. Wellbore hydraulics are modeled in Prosper while surface networks are simulated in GAP, with Python OpenServer coupling these modules to integrate real-time KPIs. The multi-objective optimization employs GPU-accelerated NSGA-II to simultaneously maximize cumulative production, extend stable production duration, and minimize total gas injection volume, subject to constraints including total gas availability, per-well production caps, reservoir pressure thresholds, and maximum injection limits. The method established in this study was used to optimize the gas injection rate of the gas lift well in one oil field in Iraq. The results show that under the same production target conditions,

the required injection volume was reduced by 84.6% after optimization, greatly saving the development cost of gas injection wells.

Key words: Gas lift optimization, proxy model, NSGA-II algorithm, Integration method

Introduction

Gas lift, as an efficient artificial lift method, reduces the liquid column density by injecting high-pressure gas into the wellbore and utilizes the expansion energy of the gas to lift well fluids to the surface^{[1][2]}. Its core advantage lies in its unique energy transfer mechanism—unlike the mechanical transmission of rod pumps or the electrical drive of electric submersible pumps (ESPs), gas lift achieves energy transfer through the dynamics of gas-liquid two-phase flow, significantly lowering the bottomhole flowing pressure and amplifying the production pressure differential^{[3][4][5]}. This characteristic makes it indispensable in drainage gas production for high-water-cut oil fields, deepwater oil and gas fields, and shale gas wells.

Compared to other artificial lift methods, although gas lift requires high upfront investment in surface pipeline networks, it offers five key advantages^{[6][7]}: 1. Lift depth exceeding 5,000 meters (e.g., Tuha Oilfield case reaching 5,327 meters); 2. Strong adaptability to complex media, handling high sand, wax, and corrosive fluids; 3. Broad liquid production range, with single-well daily liquid production up to 1,000 m³; 4. No moving downhole parts, resulting in low failure rates and simple maintenance; 5. Flexible parameter adjustment, enabling real-time response to reservoir dynamics via gas injection rate control.

In deepwater oil and gas development, gas lift demonstrates unique strengths. Statistics show that ~65% of artificial lift systems in subsea production use gas lift, especially in the North Sea, Gulf of Mexico, and Brazilian waters. This is because it overcomes inherent flaws of ESPs in deepwater environments: gas lift eliminates motor cooling and cable failure risks significantly enhance reliability. In the South China Sea's Liuhua 11-1 oilfield, 25 subsea gas lift units set a world record for collaborative operation. For shale gas development, the Fuling Shale Gas Field innovatively combined 'gas lift + foam drainage', reviving near-dead wells, restoring daily production from zero to 20,000 m³.

Current gas lift technology is evolving toward intelligence, integration, and sustainability^[8]. Tuha Oilfield established China's first intelligent gas lift management platform, enabling automatic parameter collection, real-time diagnostics, and remote gas injection control for 532 wells. This boosted lift efficiency by 10% and reduced labor costs by 20%. Hybrid lift processes (e.g., gas lift + jet pump) and rodless lift breakthroughs (e.g., electric submersible progressive cavity pumps) further expand gas lift's application boundaries, solidifying its role as the preferred solution for high-water-cut fields and complex well structures.

The optimization of gas lift systems is inherently a multi-objective decision-making problem, requiring the pursuit of an optimal solution under constraints such as minimum gas injection volume, maximum liquid production, and longest stable production duration^{[9][10]}. This process faces distinct challenges at both the single-well and well-group levels.

The core challenge lies in determining the optimal gas injection rate. Insufficient gas injection results in slug flow or churn flow within the wellbore, leading to severe liquid slippage losses and reduced lift efficiency. Conversely, excessive gas injection induces mist flow, diminishing the utilization efficiency of gas expansion energy and causing ineffective gas injection. Studies in the Nanpu Oilfield demonstrate that each gas lift well has a critical gas injection rate; beyond this threshold, liquid production growth plateaus or even declines. This critical point is dynamically influenced by factors such as reservoir pressure, water cut, and wellbore temperature profiles, which are difficult to capture accurately using traditional empirical formulas. This centers on the efficient allocation of limited gas sources among multiple wells sharing a common compressor. Changes in one well's injection rate propagate through pipeline pressure fluctuations (transmission speed: 200-300 m/s). Increased injection in one well reduces network pressure, while multiphase flow in oil pipelines alters hydraulic gradients, affecting wellhead backpressure. These interactions necessitate coordinated control to balance production across the entire well group.

This study focuses on rapid-response optimization for gas lift systems. A proxy model has been built by data driven to solve the problem that the reservoir model requires a large amount of calculation, making it difficult to couple into the optimization algorithm for repeated iterations. The well performance model for gas lift is built by Prosper, and the pipeline model is built by GAP. Then, the proxy model, Prosper model, and the GAP model were coupled in Python by using Open-server to call KPI. Finally, the multi-objective optimization algorithm is used to optimize the gas injection volume and injection pressure of every single well based on the integrated model. Through this optimization, it can greatly improve the production efficiency of gas lift wells under limited gas volume conditions, and it can release the production potential of a single well as much as possible under the oilfield management requirements.

Proxy model of reservoir inflow

Reservoir simulations are computationally intensive, especially when considering the complex physical processes involved, such as multiphase fluid flow, complex flow mechanics, and rock-fluid interactions^[11] ^[12]. Proxy models for reservoirs are simplified models used to approximate the complex behavior of full-scale reservoir simulations, providing faster calculations with reasonable accuracy^[13]. It is also essential tools in reservoir engineering, especially for scenarios where multiple simulations are required, such as indicator prediction, production optimization, sensitivity analysis, and uncertainty quantification.

The primary steps for building a proxy model include data sampling and processing, model selection, proxy modeling, model validation, and model application (as shown in Fig. 1):

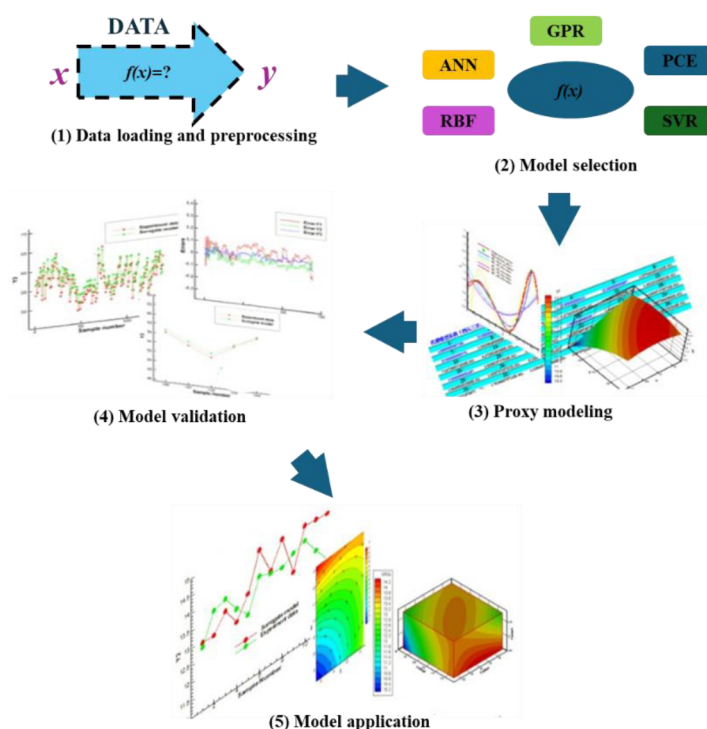


Figure 1—The specific steps of building the proxy model

Data sampling and processing: Gather input-output data from reservoir simulations, ensuring it covers a wide range of reservoir and engineer parameters. Preprocess the data, including cleaning, standardization, and normalization, to ensure high-quality data suitable for model training.

Model selection: Select an appropriate proxy modeling approach based on the data characteristics and project requirements, such as: Machine Learning Methods: Decision trees, support vector machines, random forests; Deep Learning Methods: Neural networks and convolutional neural networks, ideal for large,

complex datasets; Statistical Methods: Techniques like multiple regression or response surface methodology for simpler, linear datasets.

Proxy modeling: Train the model in the training data to establish the relationship between inputs and outputs. Optimize model parameters (e.g., regularization terms, learning rates, network structure) to ensure good fitness and generalization. Balance overfitting and underfitting, using techniques like cross-validation to achieve an optimal model.

Model validation: Evaluate the model on a test dataset using metrics such as mean squared error (MSE), root mean square error (RMSE), and R^2 to quantify performance.

Model application: Integrate the proxy model into reservoir management or production optimization systems for real-time predictions or fast decision support.

Use the model for scenario analysis and optimization, such as well placement, production parameter optimization, and injection-production ratio adjustments, enhancing reservoir development efficiency and recovery rates.

In general, there are various types of proxy models, including traditional machine learning models (GPR, SVR, RBF et al.) and deep learning models. For the purposes of applying the state of art technique to building proxy models, in this study the deep learning model is utilized to build the proxy model. In summary, proxy models are essential tools in modern reservoir engineering, providing fast, reliable approximations that help engineers make informed decisions, optimize production, and effectively manage reservoir uncertainties.

Data sampling

In proxy modeling for reservoirs, data sampling is critical, as it directly impacts the performance and reliability of the model. A well-designed sampling strategy ensures that the proxy model accurately represents the reservoir system and captures its complexity. In the data sampling step of building proxy model in this study, two classical and sampling methods called^[14]: Latin hypercube sampling (LHS) and Monte Carlo sampling (MCS) are adopted.

LHS is a statistical method used for sampling from a multidimensional distribution, particularly useful in scenarios where the goal is to explore a complex system or model with many uncertain parameters. It is a more efficient sampling technique than simple random sampling, especially when dealing with high-dimensional spaces. Monte Carlo sampling is grounded in the law of large numbers and probability theory. It works by generating a large number of random samples from a defined probability distribution to approximate the expected value of a function or the outcome of a complex process. This technique becomes more accurate as the sample size increases, based on the law of large numbers, which states that the average of results from many trials should converge to the expected value as the number of trials grows.

Model selection

After the step of data sampling, another key step of proxy modeling is model selection. In general, selecting an appropriate model can significantly improve the prediction accuracy, robustness, and computational efficiency of the proxy model. Meanwhile, there isn't a definitive theory to guide proxy model selection in reservoir simulation due to the complexity and variability of reservoir conditions, as well as the diversity of proxy models. The choice largely depends on balancing factors such as data availability, computational constraints, model interpretability, and the specific goals of the simulation. In this way, proxy model selection often involves empirical or expert to determine the most suitable model for a particular reservoir application.

Proxy modeling

The proxy model that maps KH/KV, oil density, oil rate, permult, presstran, rockcom, VRR, watervis to cumulative oil production, water cut, saturation field and pressure field. The saturation field and pressure field data are typical temporal-spatial data that change along with time, requiring careful processing and

modeling. Since the input is a vector of eight elements and the output the final saturation field and pressure field are of image data in terms of computer data structure, from a high viewpoint, we need to construct a neural network that maps one-dimensional tensor to three-dimensional tensor. At the beginning of the network, an embedding network encodes the eight parameters into three-dimensional tensors. Then, a convolutional network extracts the features more deeply. Finally, a convLSTM is imposed to output the saturation field and pressure field at final timestamp. In ConvLSTM, instead of using matrix multiplication as in traditional LSTM, convolutional operations are employed. This enables the model to capture spatial dependencies within the input data while also preserving the LSTM's capability to handle temporal sequences. The convolutional operations within the gates ensure that the model learns spatial hierarchies and captures features at different spatial scales, making it especially suitable for space-temporal tasks.

Model application

According to the steps of the proxy model, the LHS and MCS methods are utilized for sampling as the data basis of the proxy model. In the meantime, the uncertainty variables and related design spaces are defined in Table 1, while the response is defined as cumulative oil production, oil saturation field, and pressure field in this study. Noting that the other variables and responses can also be considered in the proxy model study, and in the proxy model considered the most important variables and responses of development. After the data sampling with the scale of the design space, the distribution of data samples is shown in Fig. 2.

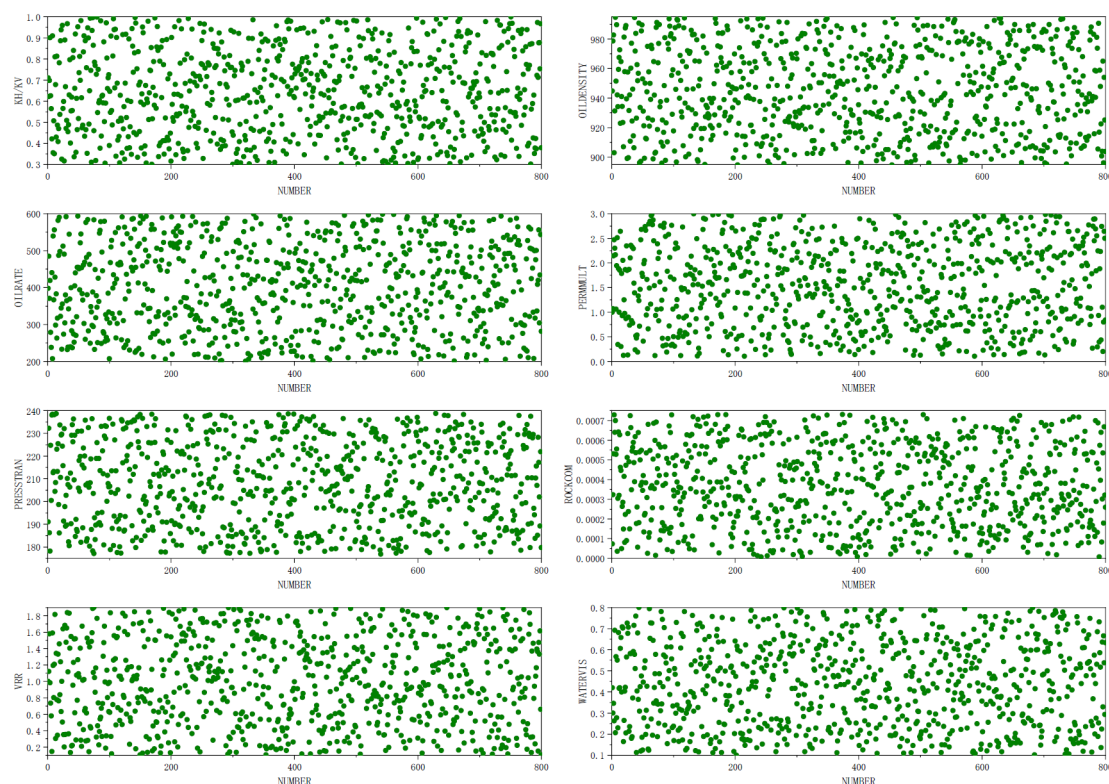


Figure 2—Distribution of data samples

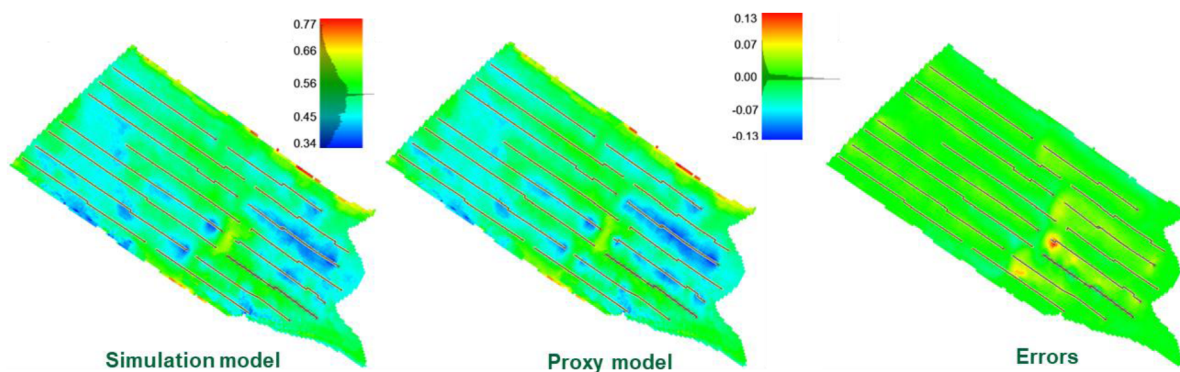
Table 1—Uncertainty variables and design spaces of #1 formation

Variable	Initial	Min	Max
KH/KV	0.71	0.3	1
OILDENSITY (kg/m ³)	944.7	894.7	994.7
OILRATE (m ³ /day)	397.5	200	600
PERMMULT	1	0.1	2
PRESSTRAN (Bar)	221.764	180.764	284
ROCKCOM(MPa ⁻¹)	7.3e-05	7.3e-06	7.3e-04
VRR	1	0.1	1.9
WATERVIS (mPa·s)	0.3	0.1	0.6

Based on the samples, the proxy model of #1 formation is established within the guide of theory in the proxy modeling section. To validate the reliability of the proxy model, the comparison between the proxy model and the numerical simulation model of the field is also conducted. The randomly generated set of variables is shown in Table 2, while the comparison of the saturation field is displayed in Fig. 3. It is obvious that the difference between the proxy model and numerical simulation model could be negligible in the field of engineering application.

Table 2—Randomly generate a set of variables to validate the accuracy of proxy model

KH/KV	OILDENSITY	OILRATE	PERMMULT	PRESSTRAN	ROCKCOM	VRR	WATERVIS
0.17	908.39	568.19	1.04	236.619	0.0004496	1.3457	0.41844

**Figure 3—Comparison of saturation field**

Well performance and ground pipeline network model

The wellbore flow models and surface pipeline network models are established using the Prosper module and GAP module of the IPM. First, based on single-well fluid data, pressure test data, etc., individual well models are built in Prosper. These models are then exported to GAP, where the surface pipeline network model is further developed based on information such as pipeline distribution and elevation. Finally, Openserver is utilized to invoke this integrated model, outputting changes in production parameters (e.g., single-well and well group production rates) under different conditions.

Well performance model in Prosper

Establishing a single-well flow model in Prosper software is a systematic engineering process requiring strict adherence to data input, model construction, and calibration procedures^[15]. The core workflow begins with the input of fundamental parameters:

PVT Data Input. In the PVT Data module, laboratory fluid analysis data must be entered, including solution gas-oil ratio (GOR), crude oil API gravity, and gas specific gravity. If flash separation test data (e.g., bubble-point pressure, oil formation volume factor, viscosity) is available, it must be supplemented. The Match All function then executes nonlinear regression to calibrate the PVT model against experimental measurements.

IPR Model Construction. The reservoir inflow performance relationship (IPR) model is built by selecting theoretical models (e.g., Vogel, Darcy, Jones) based on reservoir type. Key static parameters (e.g., permeability, reservoir thickness, skin factor) are input to characterize formation deliverability.

Wellbore Configuration.

1. Well Trajectory: Inclination survey data (measured depth vs. true vertical depth) is imported to generate a 3D well path.
2. Tubing/Casing Design: Tubing/casing sizes, downhole tools (e.g., safety valves, chokes), and abrupt diameter changes are input segmentally. Friction coefficients are configured based on pipe material properties.
3. Thermal Profile: Geothermal gradients require surface temperature and reservoir midpoint temperature. Prosper predicts temperature profiles using linear or exponential models. If downhole temperature data exists, the total heat transfer coefficient (U-value) is inverted to correct the thermal model.

VLP/IPR Calibration. This critical validation step involves:

1. Inputting ≥ 3 downhole pressure gauge measurements (production rate, flowing pressure, GOR, water cut, and gauge depth with $<10\text{m}$ error) in the Matching module.
2. Using Correlation Comparison to screen multiphase flow models (e.g., Hagedorn-Brown, PE4, Gray), ensuring test points lie within the Fancher-Brown (lower bound) and Duns-Ros (upper bound) envelopes.
3. Selecting a base correlation (e.g., PE4) and performing nonlinear regression to apply:
Gravity multiplier (γ): Adjusts fluid density gradients (typical range: 0.95-1.05).
Friction multiplier (λ): Corrects for pipe wall roughness effects (typical range: 0.85-1.15).

pipeline network model in GAP

Building a surface pipeline network model in GAP software is a systematic engineering process integrating reservoir dynamics, wellbore flow, and surface gathering systems, with its core focus on coupling the entire production system^{[16][17]}. The initial modeling phase requires consolidating multi-source data:

Import pipeline topology composed of nodes (wellheads, separators, compressors) and pipelines (pipes, valves), defining parameters including pipe diameter, length, elevation change, and material properties (e.g., friction factor: 0.015-0.025 for galvanized steel pipes, 0.005-0.015 for coated pipes).

Input fluid composition data (oil/gas/water properties) and calibrate PVT parameters through nonlinear regression.

Incorporate ground elevation data to ensure pipeline burial depth and slope align with actual terrain.

Validate topological consistency via dual verification methods:

Geometric Networking: Determine connections based on coordinated distances.

Attribute Networking: Establish associations through pipeline identifiers.

Wellbore-Network Integration

Achieve coordination between wellbores and the pipeline network by invoking PROSPER's single-well models to simulate interactions within gas injection and oil production systems.

NSGA-II algorithm

NSGA-II (Non-dominated Sorting Genetic Algorithm II) stands as a landmark algorithm in the field of multi-objective optimization^[18]. Its core value lies in efficiently solving complex problems with conflicting objectives (e.g., simultaneously minimizing cost and maximizing performance). Through three key innovations—Fast Non-dominated Sorting, Crowding Distance Mechanism, and Elitism—it significantly enhances the solution efficiency of the Pareto Front and the distribution quality of the solution set.

Fast Non-dominated Sorting is central to NSGA-II's hierarchical solution screening. It stratifies individuals in the population into multi-level Fronts based on dominance relationships: Front 1 contains all solutions not dominated by any other individuals. Subsequent fronts are recursively generated by removing already stratified individuals. This process reduces computational complexity from $O(MN^3)$ in traditional NSGA to $O(MN^2)$ (M = number of objectives, N = population size), dramatically accelerating large-scale problem solving^[19]. The flow chart of the NSGA-II is shown in Fig. 4. Crowding Distance maintains solution diversity by quantifying the distribution density of adjacent solutions in the objective space. The metric is calculated as:

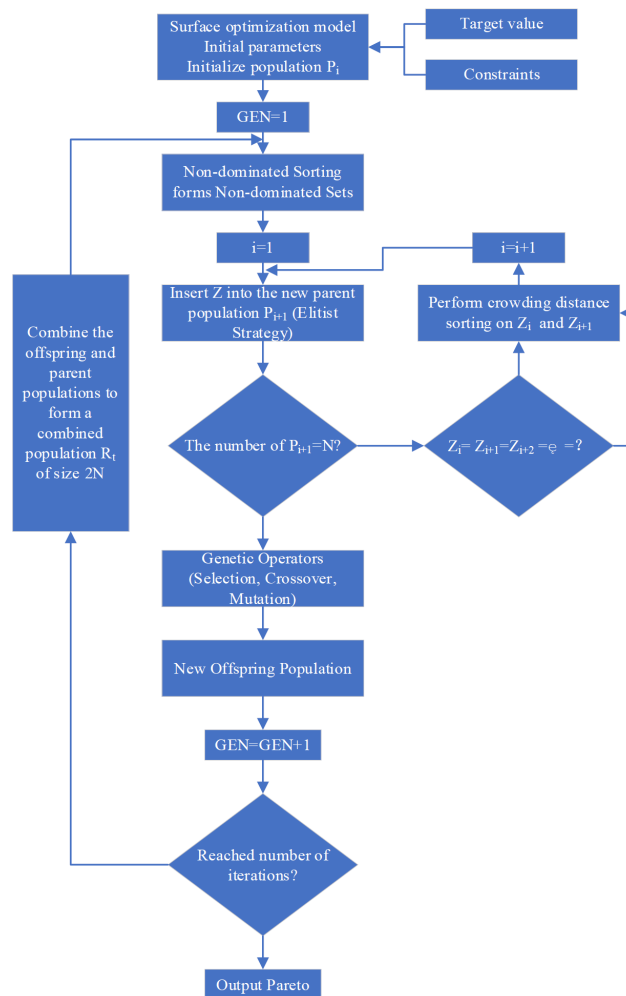


Figure 4—Flow chart of NSGA-II

$$Ci = \sum_{m=1}^M \frac{d_i^m}{r_{max}^m - r_{min}^m}$$

where: d_i is Euclidean distance between solution i and its nearest neighbors along objective m ; r_{max} , r_{min} is Maximum and minimum values of objective m in the current front; Boundary solutions (those with extreme objective values) are assigned infinite crowding distance to preserve solution spread.

NSGA-II's technical breakthroughs manifest in three key dimensions^[20]:

1. Efficiency & Robustness

Compared to traditional NSGA (Table 3), its optimized computational complexity combined with elitist strategy enables handling high-dimensional objectives and nonlinear constraints.

2. Solution Set Quality

The crowding distance mechanism ensures uniform distribution across the Pareto front, comprehensively covering the objective space. In benchmark cases (e.g., ZDT1 function tests), its solution distribution significantly outperforms algorithms lacking this mechanism.

3. Application Universality

From engineering optimization (e.g., balancing lift vs. drag in aircraft wing design) to financial portfolios (return-risk tradeoffs), NSGA-II consistently provides diverse compromise solutions, empowering decision-makers to select based on preference priorities.

Table 3—Key Performance Comparison between NSGA-II and Traditional NSGA

Characteristics	NSGA	NSGA-II
Computational Complexity	$O(MN^3)$	$O(MN^2)$
Diversity Maintenance	sharing radius σ_{share} (manual tuning required)	Crowding distance (self-adaptive)
Elitism Preservation	Not Supported	Supported

NSGA-II establishes the benchmark framework for multi-objective optimization algorithms through the synergistic innovation of three core mechanisms: rapid hierarchical solution screening, adaptive distribution control, and historical solution reuse. Its technical superiority manifests not only in enhanced computational efficiency and solution set quality, but also in versatile adaptation to complex cross-domain demands spanning engineering, finance, and beyond.

Integration model of optimization

The flow chart of the integration model of optimization is shown in Fig. 5. This model including 4 important unit:

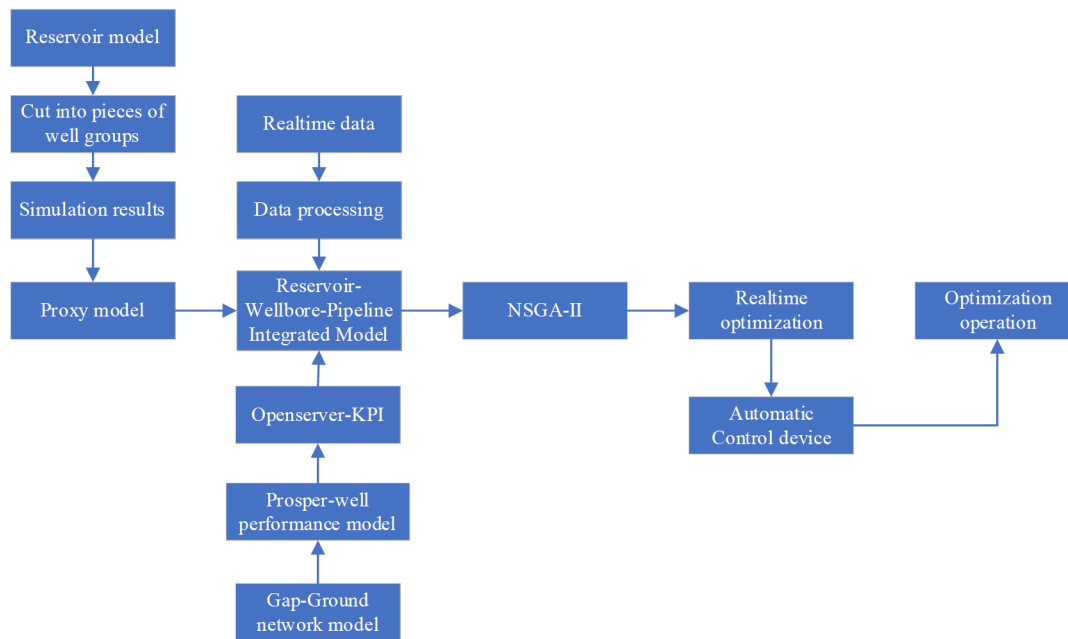


Figure 5—Flow chart of Integration model

Reservoir Model

The reservoir model serves as the foundation of the entire system, simulating subsurface hydrocarbon reservoirs with geological structures, lithology, and fluid properties. This model is segmented into well groups for granular analysis of distinct production units. Each segmented well group undergoes independent simulation, generating dynamic production data (e.g., pressure, flow rate). Simulation results feed into constructing a proxy model—a simplified surrogate (e.g., machine learning model) for rapid prediction of production responses under varying operational conditions. Field-acquired real-time data (wellhead pressure, temperature, flow rate) undergoes data processing (cleaning, calibration, formatting) to ensure quality before integration.

Integrated Model Coupling

Commercial software enables integration via the IPM suite: GAP-Models surface network pressure distribution and optimizes system-wide coupling (reservoir-wellbore-pipeline). Prosper-Focuses on single-well optimization, simulating wellbore fluid dynamics and lift efficiency to calculate wellhead pressure under varying bottom-hole conditions. Openserver-Acts as the integration hub, enabling cross-module data exchange and automated scripting. Coupling Logic-Prosper computes wellhead pressure & GAP calculates surface pressure distribution. Pressure/flowrate coupling at the wellhead achieves reservoir-wellbore-surface separator integration. Python invokes keywords by IPM Openserver to orchestrate GAP/ Prosper, forming a closed-loop system.

Multi-objective Optimization (NSGA-II Core Algorithm)

The integrated model is wrapped with NSGA-II to resolve conflicting objectives while extending stable production duration. The key inputs are the proxy model for rapid prediction, the integrated model for high-fidelity simulation, the openserver-KPI module for constraints (e.g., recovery factor, cost). The Output is the Pareto optimal solution set (tradeoff solutions).

Real-time Optimization & Automated Control

Dynamic Parameter Adjustment: NSGA-II outputs feed into real-time optimization, dynamically constraining gas injection volume/pressure based on field conditions. Singular Operating Point Selection: Recommends unique parameters from the Pareto set for field implementation. Automated Execution: Automatic control devices (e.g., gas lift valves) adjust operations using recommended setpoints. Closed-loop Learning: Each optimization cycle is recorded to refine subsequent decisions (optimization operation).

Result

The target field in Iraq has entered the late production phase with declining reservoir pressure. The block comprises three well groups, each containing varying numbers of flowing wells. Within Group 3, Wells #3-1, #3-2, and #3-3 are planned for conversion to gas lift production due to productivity deterioration. The integrated optimization approach in this study was applied to conduct feasibility analysis for gas lift implementation and optimize gas injection volume and liquid production rate. In this way, can achieve dual objectives: minimizing total gas injection while maximizing oil production.

The process of the three well groups and surface pipeline network in GAP is shown in [Fig. 6](#). Group 1 goes directly to the separator, while Group 2 and Group 3 converge into a pipeline and enter the separator. Among them, there are 12 natural flowing wells in Group 1, 8 natural flowing wells in Group 2, and 7 natural flowing wells in Group 3, totaling 27 wells, all of which produce the same layer. Among them, the wells expected to be converted to gas lift production are # 3-1, # 3-2, and # 3-3 in Group 3.

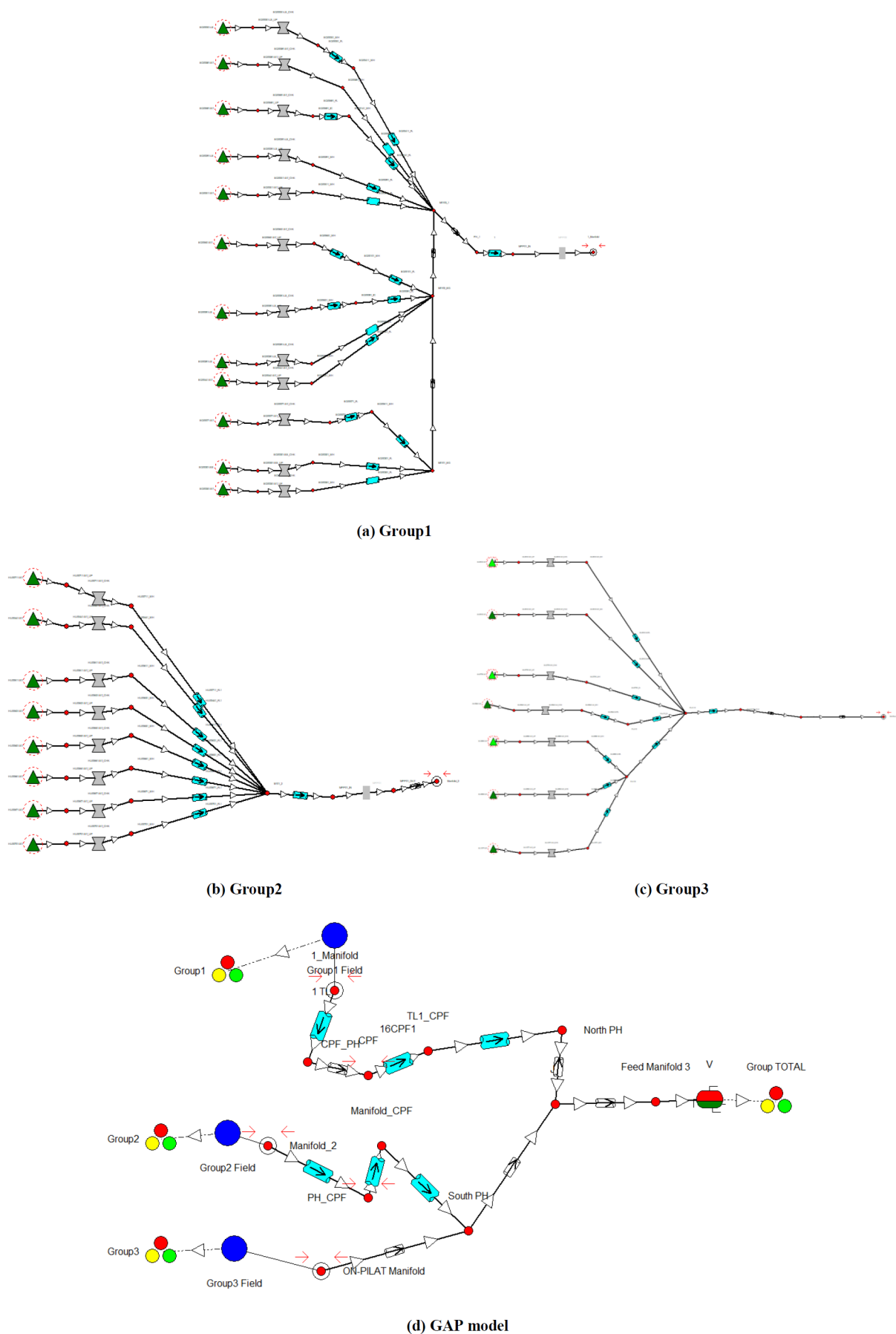


Figure 6—Model of various well groups in GAP: (a) Group1; (b) Group 2; (c) Group 3; (d) GAP model.

According to the actual situation of the ground, the current maximum gas injection rate is 10 MMscf/day, which may can be expanded to 50 MMscf/day in the next two years. Two cases were simulated: Case 0 with a fixed gas injection rate of 2 MMscf/d for all three wells, and Case 1 with algorithm-optimized gas injection rates for each well. From the results, the cumulative production for both cases is nearly identical, as shown in the Fig. 7. However, the optimized case (Case 1) requires significantly less gas injection: Case0 consumes 2,340 MMscf/year, while Case 1 uses only 358.55 MMscf/year—an 84.7% reduction—drastically lowering gas injection costs, as shown in Fig. 8. The monthly injection rates after optimization (detailed in the Table 4) show consistent rates for Wells #3-1 and #3-3, while Well #3-2 starts with lower injection in the initial phase but aligns with #3-1 and #3-3 in later phases.

Table 4—Gas injection rate

Time (date)	#1(MMscf/d)	#2(MMscf/d)	#3(MMscf/d)
2025/4/13	0.299	0.198	0.299
2025/5/13	0.299	0.198	0.299
2025/6/13	0.299	0.198	0.299
2025/7/13	0.299	0.198	0.299
2025/8/13	0.299	0.198	0.299
2025/9/13	0.299	0.198	0.299
2025/10/13	0.299	0.198	0.299
2025/11/13	0.299	0.198	0.299
2025/12/13	0.299	0.198	0.299
2026/1/13	0.299	0.200	0.299
2026/2/13	0.499	0.472	0.499
2026/3/13	0.488	0.488	0.488
2026/4/13	0.352	0.352	0.352

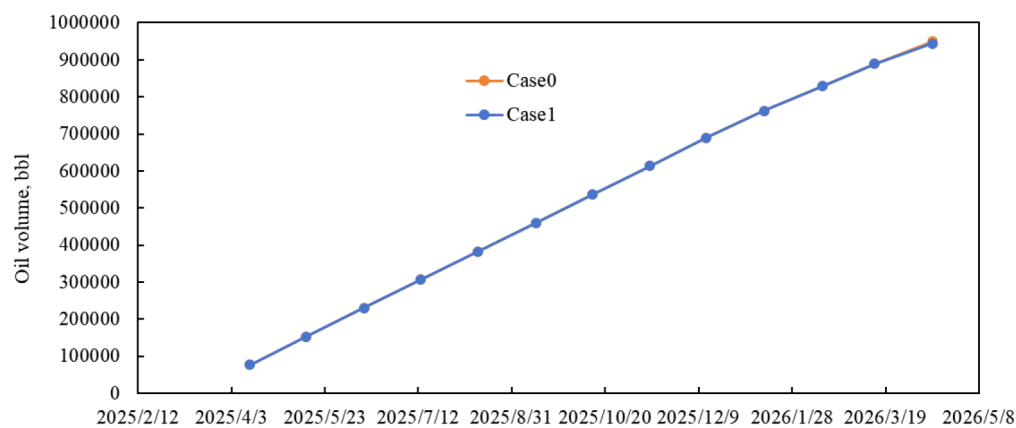


Figure 7—Cumulative gas injection volume

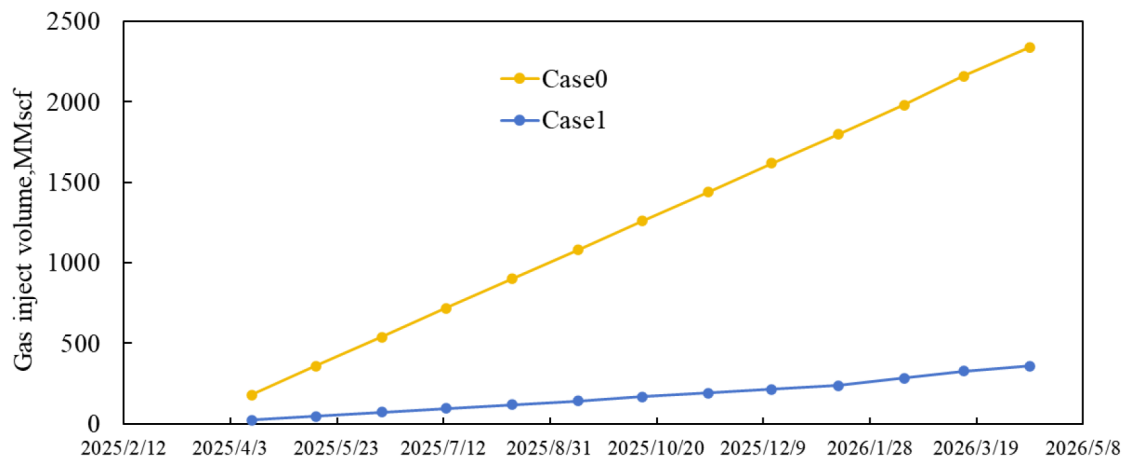


Figure 8—Cumulative gas injection volume

Conclusion

This study presents an integrated reservoir-wellbore-pipeline optimization framework for gas lift systems, combining deep learning proxy modeling, GPU-accelerated NSGA-II, and real-time control strategy to address the multi-objective challenges of gas injection rate allocation under limited supply. The key contributions and findings are summarized as follows:

1. A ConvLSTM-based proxy model achieves acceleration in reservoir simulation while maintaining <10% prediction error compared to full-physics models (validated against saturation/pressure field outputs). This resolves computational bottlenecks for iterative optimization.
2. GPU-parallelized NSGA-II reduces optimization time through distributed fitness evaluation and non-dominated sorting, enabling rapid Pareto front generation for complex well groups.
3. Through case verification analysis, it is found that using the method in this study can reduce the gas injection volume by 84.6% while ensuring that the production rate remains basically unchanged.

Acknowledgement

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